EXPERT VOICES

Global majority countries must embed critical minerals into AI governance

CHINASA T. OKOLO Authors Info & Affiliations

SCIENCE • 5 Mar 2026 • Vol 391, Issue 6789

Over the past decade, an increasing number of Global Majority countries—broadly characterized as economically developing countries in Africa, the Caribbean, Asia, Latin America, the Middle East, and Oceania—have introduced national artificial intelligence (AI) strategies. However, many of these frameworks overlook a critical dimension of AI governance: the strategic importance of critical minerals. As geopolitical tensions intensify around access to computing infrastructure, Global Majority countries must recognize that their mineral reserves represent leverage points for transforming their respective positions in the global AI value chain.

Graphical processing units (GPUs) and semiconductors, the computational backbone of AI, depend on an array of critical minerals, such as cobalt, nickel, copper, lithium, tungsten, and rare earth elements. Alongside China, which dominates rare earth processing, Global Majority countries serve disproportionately as extraction sites for these resources.

Mineral extraction generates severe environmental degradation and widespread human rights violations. Mining operations contaminate water supplies, displace communities, and operate under conditions that frequently involve forced labor, child labor, and dangerous working environments. These harms are compounded by structural inequalities, such as disparities in advanced education and digital infrastructure, that position Global Majority countries as suppliers of raw materials and data annotation labor rather than as developers of high-value AI models and applications. The result is a deeply uneven value chain.

While semiconductors, GPUs, and advanced computing infrastructure represent the most profitable segments of the AI economy, Global Majority countries are locked out of these sectors. Their workers are relegated to low-wage tasks such as data labeling and content moderation; work that is fraught with economic exploitation and mental health harms, yet essential to training the AI systems that generate wealth elsewhere.

Critical mineral extraction has long served as a site for the exercise of geopolitical power, and the AI race adds new urgency to these dynamics. Russia has established extensive mining partnerships throughout Africa and Latin America, partnerships that complement China's Belt and Road Initiative and mineral-acquisition efforts by middle powers such as the United Arab Emirates.

The United States has similarly intensified its pursuit of access to critical minerals. In December 2025, the U.S. Department of State formed Pax Silica, a nine-member strategic coalition that also included Japan, South Korea, Singapore, the Netherlands, the United Kingdom, Israel, the United Arab Emirates, and Australia. The coalition's stated purpose is to secure AI, semiconductor, and critical mineral supply chains while reducing dependency on China. No Global Majority countries were included as members at the outset, although India has since joined the coalition. This exclusion sends a clear message that the United States views these nations as resource suppliers rather than as equal partners in the global distribution of critical minerals.

In February 2026, the U.S. hosted the Critical Minerals Ministerial and signed partnerships with Argentina, the Cook Islands, Ecuador, Guinea, Morocco, Paraguay, Peru, the Philippines, and Uzbekistan. An earlier agreement with the Democratic Republic of Congo (DRC) promised substantial investments and security guarantees in exchange for prioritized access to cobalt reserves. This partnership reflects a fundamentally unequal power dynamic: the DRC possesses the world's largest cobalt reserves while simultaneously grappling with extreme poverty, labor exploitation, food insecurity, displacement, and violent conflict.

Recognizing power imbalances in critical mineral extraction, regional bodies have begun developing frameworks to assert greater control over critical mineral governance. In February 2025, the African Union adopted Africa's Green Minerals Strategy, a continental framework designed to transform Africa from a raw material exporter into an integrated partner in global clean technology value chains. The strategy emphasizes local processing, environmental sustainability, and value addition before export.

Similarly, the Association of Southeast Asian Nations (ASEAN) has established the ASEAN Minerals Cooperation Action Plan, a five-year blueprint aiming to position Southeast Asia as a competitive, sustainable, and transparent hub for mineral production and processing. While the Economic Commission for Latin America and the Caribbean (ECLAC) lacks a formal critical minerals strategy, it has articulated the need to leverage the region's vast lithium, copper, and nickel reserves to drive economic development, promote local processing and industrialization, and support the transition to cleaner energy systems.

These regional initiatives represent important steps toward collective bargaining power. However, their success depends on moving beyond aspirational declarations toward sustained, enforceable commitments. Global Majority countries must translate these frameworks into concrete policies that prioritize local capacity building, technology transfer, and equitable partnerships.

India offers a potential model for how Global Majority countries can leverage critical mineral reserves and skilled labor to move up the AI value chain. The country possesses large reserves of bauxite, chromite, and manganese, alongside one of the world's largest bases of technical talent. This combination, along with India's status as a global technology hub and an entry point for digital technologies in emerging markets, provides a foundation for broader industrialization efforts. With companies like Apple expanding phone manufacturing operations to India and Vietnam, chip manufacturing represents a logical next step—one that could catalyze domestic AI development and cement India as an AI leader amongst Global Majority countries.

However, the AI Impact Summit, held in February 2026 in New Delhi and noted as the first event in the Global AI Summit series hosted by a non-Western country, reveals a critical gap. Despite the significance of the convening, the summit included few sessions addressing critical minerals or strategies for leveraging these resources to build capacity for semiconductor manufacturing and AI development. While previous summits have produced numerous broad declarations and ephemeral commitments to AI cooperation, the continued absence of substantive discussion on critical minerals underscores the need for Global Majority countries to move beyond symbolic participation toward asserting genuine influence over AI governance.

This will require moving beyond reactive extraction agreements toward proactive industrial policies that prioritize several key principles. First, these countries must expand local processing and refining capabilities. To advance this goal, countries such as Indonesia, Namibia, and Zimbabwe have recently banned the export of raw materials. Exporting raw materials perpetuates dependency, while adding value domestically through processing creates jobs, builds technical expertise, and captures a greater share of economic returns. Second, any partnerships with foreign powers must include genuine technology transfer and capacity building, not merely capital investment in exchange for resource access. Third, Global Majority governments must invest in the foundational infrastructure necessary for participating in advanced manufacturing and AI development, such as digital connectivity; science, technology, engineering, and mathematics (STEM) education, and research institutions.

These efforts must be accompanied by stringent regulation of environmental and labor conditions in mining operations. While many countries have existing laws that mandate worker protections and compliance with environmental protection laws, regulatory loopholes enable illegal mining, necessitating frequent legal reforms, which are underway in countries such as Ghana and Chile. The human and ecological costs of critical mineral extraction can no longer be treated as externalities in the pursuit of economic development and AI advancement. Sustainable industrialization requires protecting the communities and ecosystems most affected by extraction.

Critically, Global Majority countries should prioritize South-South cooperation. Such work is already underway through efforts such as the Council for Critical Minerals Development in the Global South and a new partnership between South Africa and the DRC that aims to increase co-development of projects, technology transfer, and policy advocacy. By sharing resources, expertise, and governance frameworks, Global Majority nations can reduce their dependence on extractive partnerships with Western and other powers. When global coalitions exclude these countries, regional alliances provide the institutional infrastructure to negotiate from positions of strength rather than dependency.

Strategic critical minerals partnerships, when structured equitably, can serve as catalysts for sustainable industrialization, technological capacity building, and genuine participation in the AI value chain. But this transformation will not happen automatically. Without asserting sovereignty over their critical mineral resources, Global Majority countries will remain locked in an exploitative relationship to AI development, providing the materials and labor that power the technology while frontier AI powers capture its economic rewards and governance authority.